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CONNECTION DESIGN ADVISOR

Project: Quick Analysis

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Of course. As STRUCTMIND AI, I will provide optimal connection recommendations for your 3-story SMF + OCBF mixed system. The design philosophy for each system differs significantly: SMF connections are designed for ductility and energy dissipation via controlled yielding, while OCBF connections are designed to remain elastic and be stronger than the yielding/buckling braces.

Here is the breakdown for each connection type:

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1. SMF Beam-to-Column Joints (Interior & Exterior)

• Recommended Connection Type: Reduced Beam Section (RBS) Moment Connection. Commonly known as a "dogbone," this is a prequalified connection per AISC 358-22. It is robust, well-researched, and cost-effective. The connection forces a plastic hinge to form in the beam, away from the column face, protecting the column and the beam-to-column weld.

• Governing Limit States (AISC 341-22 E3 & AISC 360-22 J10):

The connection is designed using capacity design principles, where it must resist the forces generated by the beam reaching its expected plastic moment capacity at the RBS (M_{pr}). Strength of the Groove Weld: The CJP groove weld connecting the beam flanges to the column flange must develop the strength of the beam flange. Shear Plate/Tab Connection: Bolt Shear Rupture and Bearing on the beam web and plate. Gross Yielding, Net Rupture, and Block Shear Rupture of the shear plate. Weld strength (plate-to-column). Column Side Limit States (AISC 341-22 E3.6f): Column Panel Zone Shear Yielding. Column Flange Local Bending. * Column Web Local Crippling.

• Bolt / Weld Sizing Guidance:

• Welds: Use CJP groove welds for beam flange to column flange connection. Fillet welds for shear plate to column must be designed to resist the amplified shear force, V_p, associated with the formation of plastic hinges.

• Bolts: High-strength bolts (e.g., ASTM F3125 Gr. A325 or A490) connecting the shear plate to the beam web must also be designed for the amplified shear, V_p.

• Constructability Score: 8/10

• Remarks: Excellent. The RBS cut is performed in the shop under controlled conditions. Field erection involves only bolting the shear plate and is relatively tolerant. It avoids complex field welding.

• Alternate Solutions:

• Bolted Unstiffened/Stiffened Extended End Plate (BUEEP/BSEEP): Another prequalified connection. Can be faster to erect as it's a fully bolted field connection. Requires tighter fabrication and erection tolerances.

• Bolted Flange Plate (BFP): Good for retrofit or when welding to the column flange is not desirable. More pieces to handle in the field.

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2. OCBF Brace-to-Gusset Connection

• Recommended Connection Type: Gusset Plate with Direct Bolted or Welded Interface. For HSS braces, this often involves a slotted HSS with fillet welds to the gusset. For W-shape braces, bolts are placed through the flange or web.

• Governing Limit States (AISC 341-22 F1.6 & AISC 360-22 J4):

The connection must be designed to be stronger than the brace. The required strength is the lesser of the expected brace tensile strength ($R_t A_g F_y$) and the maximum force effect that can be delivered by the system. Brace to Gusset: Tensile Yielding and Rupture of the brace member. Block Shear Rupture on the brace at the bolt line. Bolt Shear Rupture and Bearing. Weld Rupture (for welded connections). Gusset Plate: Tensile Yielding on the Whitmore effective width. Shear Yielding/Rupture along the interface with the beam/column. Gusset Plate Buckling (compression). Gusset to Frame: Strength of welds or bolts connecting the gusset to the beam/column.

• Bolt / Weld Sizing Guidance:

• Use the Uniform Force Method (UFM) to properly model force flow and design the gusset plate and its interfaces.

• Provide a clearance of at least two times the gusset thickness between the end of the brace and the beam/column to allow for inelastic rotation.

• Bolts should be designed for the required brace strength, often slip-critical in the final condition.

• Constructability Score: 7/10

• Remarks: Gusset plate geometry can be complex, requiring careful detailing and fabrication. Field fit-up, especially at skewed intersections, can be challenging. Bolted connections are generally preferred over field welding for speed and quality control.

• Alternate Solutions:

• Proprietary Cast Connectors (e.g., Cast Connex): These can simplify the gusset design, improve performance, and provide a cleaner aesthetic, but come at a premium cost.

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3. Column Base Plates

- Recommended Connection Type: Exposed Column Base Plate with Anchor Rods. The design philosophy will depend on which system governs the column.
- For SMF Columns: A Fixed Base is required. It must be designed to develop the plastic moment capacity of the column (M_{pc}) to ensure yielding occurs in the beam hinges above. This requires a thick base plate and robust anchoring.
- For OCBF-only Columns: A Pinned Base is often sufficient, designed to transfer only axial load and shear.
- Governing Limit States (AISC 360-22 J8 & AISC 341-22 D2.6):
- Base Plate: Flexural Yielding of the plate under bending.
- Concrete & Anchors (per ACI 318):
- Concrete Bearing Strength.
- Anchor Rod Tensile and Shear Strength.
- Concrete Tensile Breakout Strength.
- Concrete Shear Breakout Strength.
- Concrete Pryout Strength.
- Side-Face Blowout.
- Column: Weld strength of column to base plate.
- Bolt / Weld Sizing Guidance:
- For SMF fixed bases, design for the required moment, M_f, specified in AISC 341 D2.6. This will result in large anchor rods (often 1.5" to 2.5" dia.) and thick plates (2" to 4").
- For pinned bases, design for factored axial compression and shear. Minimize plate thickness to ensure it doesn't attract significant moment.
- Use leveling nuts for erection tolerance and ensure proper non-shrink grout installation.
- Constructability Score: 6/10
- Remarks: Critical path activity. Anchor rod placement is prone to error and requires careful survey and quality control. Handling heavy plates and coordinating with concrete trades adds complexity.
- Alternate Solutions:
- Embedded Column Base: Can provide excellent fixity but is more complex to design and construct. Not typical for a 3-story building.
- Shear Key: A steel element embedded in the footing to transfer large shear forces, reducing the demand on anchor rods.

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4. OCBF Beam-to-Column Connection

• Recommended Connection Type: Bolted Double Angle or Shear Tab (Single Plate) Connection. This is a standard simple shear connection. However, it must also be checked for axial loads transferred from the brace-gusset assembly.

• Governing Limit States (AISC 360-22 J2, J4, J5):

• Bolt Shear Rupture and Bearing.

• Shear Yielding and Rupture of the connecting element (angle or plate).

• Flexural Rupture of the connecting element.

• Block Shear Rupture on the beam web and connecting element.

• Weld strength (if shop-welded to the column).

• Crucial Check (AISC 341-22 F1.6c): The beam and its connections must be designed for the combined effects of gravity loads plus the axial forces generated by the unbalanced brace forces.

• Bolt / Weld Sizing Guidance:

• Size bolts and welds for the combination of gravity shear and the axial force demand from the brace system analysis.

• Use standard hole sizes and typical detailing for simple shear connections to allow for rotation.

• Constructability Score: 9/10

• Remarks: Extremely common, fast, and forgiving. Erectors are highly familiar with these connections. They provide good erection tolerance.

• Alternate Solutions:

• Single Angle Connection: Can be used for lighter loads, but double angles provide more stability during erection.

• Bolted End Plate Shear Connection: Can be faster if fabricated with automation, but requires tighter longitudinal tolerance.

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5. Column & Beam Splices

• Recommended Connection Type: Bolted Flange Plate Splice.

• Column Splice: Located ~4 ft above the floor level for ease of access. This connection uses plates on both sides of the flanges and possibly the web to transfer all applicable forces.

• Beam Splice (less common): If required, it must be located outside the protected zones (i.e., away from the RBS in an SMF beam).

• Governing Limit States (AISC 360-22 J6 & AISC 341-22 D2.5):

• For SMF columns: The splice must be designed to develop at least 50% of the plastic flexural strength (M_{pc}) of the smaller column section. If the column is part of the OCBF and subject to net tension, it must develop the required tensile strength.

• General Limit States:

• Bolt Shear Rupture and Bearing.

• Gross Yielding and Net Rupture of the splice plates.

• Block Shear Rupture of the splice plates and connected members.

• Bolt / Weld Sizing Guidance:

• Size bolts and plates based on the force demands. For SMF columns, this will be the seismic requirement from AISC 341. For OCBF columns, it will be the factored loads from analysis.

• Use slip-critical bolts if slip at the splice would be detrimental to frame performance.

• Constructability Score: 8/10

• Remarks: A very practical and erection-friendly solution. Allows columns to be fabricated and shipped in manageable lengths (e.g., two stories). Proper alignment during erection is key.

• Alternate Solutions:

• CJP Groove Weld Splice: Provides a seamless, full-strength connection but is extremely expensive and slow for field work. It requires highly skilled labor, special inspection (NDT), and is sensitive to weather. Generally avoided unless required for architectural or extreme load transfer reasons.